

Vishesh Singh

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Professional Summary

Entry-level Data Scientist and Machine Learning Engineer with strong hands-on experience in Python, ML, NLP, and end-to-end AI system development. Proven ability to build, deploy, and monitor data-driven applications using modern MLOps practices. Actively seeking data science or ML roles where analytical rigor meets real-world impact.

Technical Skills

Programming: Python, C++

Databases: MySQL

Data Science & ML: Machine Learning, Deep Learning, Natural Language Processing, Exploratory Data Analysis, Feature Engineering, Model Evaluation, Model Deployment

MLOps: DVC, MLflow, Weights & Biases, Docker, GitHub Actions

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch

Deployment & APIs: FastAPI, Flask, Streamlit, Hugging Face Transformers

Generative AI: LLM Integration, LangChain, RAG, Vector Stores, Document Loaders

Projects

AI Restaurant Assistant

[Github](#)

- Generated realistic **restaurant datasets** using **Faker** for system testing and validation.
- Designed **modular backend services** for **booking**, **ordering**, and **reviews** using **Python** and **SQLAlchemy**.
- Implemented a **LangChain ReAct agent** powered by **Gemini** for **natural language-driven operations**.
- Built a **Streamlit-based chat interface** for **real-time restaurant management**.

Real-Time BBC News Summarization & Sentiment Analysis System

[Github](#)

- Built an automated **news scraping pipeline** using **BeautifulSoup** for live article ingestion.
- Integrated **GROQ API** for **abstractive summarization** and **sentiment analysis**.
- Implemented **text-to-speech (TTS)** output using **Google TTS**.
- Deployed **FastAPI backend services** with a **Streamlit dashboard** for **interactive analytics**.

BERT Fine-Tuning for Sentiment Analysis

[Github](#)

- Fine-tuned a **BERT-based model** using **Hugging Face Transformers**, achieving **93.43% accuracy**.
- Integrated **Weights & Biases (W&B)** for **experiment tracking** and **performance monitoring**.
- Built an optimized **real-time inference pipeline** for **sentiment prediction**.
- Improved **training throughput** to **23.32 samples/sec** through optimization techniques.

Real Estate Price Prediction, Recommendation & Analytics

[Github](#)

- Built a **machine learning system** to **predict real estate prices** with an **R² score of 0.99**.
- Performed **data cleaning**, **feature engineering**, and **model evaluation** using **Scikit-learn** and **Pandas**.
- Implemented **DVC** for **dataset versioning** and **MLflow** for **experiment tracking**.
- Deployed the **analytics dashboard** using **Streamlit**.

YouTube Sentiment Analysis Plugin

- Developed a **Chrome extension** for **real-time sentiment analysis** of **YouTube comments** using **NLP** and **Scikit-learn** models.
- Built automated **text preprocessing pipelines** (tokenization, stop-word removal) with reproducibility using **DVC**.
- Tracked experiments, metrics, and model versions using **MLflow** for scalable monitoring.
- Integrated the trained **ML inference pipeline** directly into the browser extension for seamless inference.

Education

Bachelor of Technology (B.Tech) in Computer Science & Engineering

CSJM University, Kanpur

Nov 2021 – June 2025

Certifications

- ML & Data Science Program (DSMP 1.0 & DSMP 2.0) – CampusX
- Machine Learning Master Program – Euron